

Aviation and Missile
Research, Development and Engineering Center
Engineering Directorate
Quality Management Division

ORIENTATION ON
FIRST ARTICLE INSPECTION (FAI)
(CONTRACTOR TESTING)

REDSTONE ARSENAL, ALABAMA

Revised 1/26/2017

ORIENTATION ON FIRST ARTICLE INSPECTION (FAI)

INTRODUCTION

This document is provided as an orientation to familiarize you with the purpose and scope of first article inspection (contractor testing) as performed on this contract. It is a more detailed review of the processes, controls and procedures described in the first article inspection and test clauses of the contract, FAR 52.209-3 and 52.209-4020. Moreover, it is provided for informational purposes only and is not part of the terms and/or conditions of the contract. The contents of this orientation document shall not be construed or interpreted as direction to perform tasks, work, or labor which could result in a monetary charge to the U.S. Government. Additionally, the contents of this document shall not be used as direction to modify any existing contract between vendors and the U.S. Government. For questions regarding contracts and changes to contracts, consult your contracting officer. The contracting officer is the approving authority for all matters related to contracts.

SCOPE

Purpose. The purpose of the first article inspection and approval process is to ensure that the Contractor has the capability to manufacture a product that conforms to all contract requirements. The Contractor must notify the Procurement Contracting Officer (PCO) and the Government surveillance element of the contractual scheduled date, time, and location for performance of the required FAI, in writing, no less than 30 days prior to the scheduled first article inspection and test date. Notification must include the date, time and location of scheduled testing. Participation in the first article inspection and test is an option available to the Aviation Missile Research, Development, and Engineering Center (AMRDEC) Quality Management Division. In addition to Quality Management personnel, other Command representatives may also participate. Likewise, it is imperative that your local Defense Contract Management Agency (DCMA) Quality Assurance Representative (QAR) participate in the inspection and/or tests, whether or not AMCOM/AMRDEC representatives attend. The on-site inspection should communicate expectations for the First Article Test Report (FATR) so it will be determined acceptable on first submission, thus preventing delays in approval. In the event that there is no purchasing Command participation, following the process in this Orientation should maximize the likelihood of a successful FAI.

Meeting. A thorough review of the procurement package is accomplished to determine the technical scope involved and to insure that concerned personnel are intimately familiar with overall requirements. The FAI and test process normally begins with a meeting between all cognizant Government and Contractor personnel. The purpose is to review the scope of the first article inspection and test, and to agree upon the manner in which it will be conducted. At this time, a Contractor representative is normally designated as a focal point of contact for Government participants to coordinate actions and to obtain authoritative responses to matters that may arise. Meetings are held as often as necessary, and a final critique is conducted upon completion of the inspection. The FAI process will **neither exceed nor modify** the contract requirements in any way. Any contract modifications, if required, will be through the Contracting Officer(s).

Inspection.

The inspection and test of the first article candidate is the responsibility of the Contractor. However, the Government representatives may, at their option, perform any of the required inspections and/or tests with assistance from any specially trained Contractor personnel necessary to operate unique or complex special inspection and test equipment. The on-site first article inspection consists of, but is not limited to, a thorough examination of certifications, employee training records (where certifications are required), material, processes, procedures, item characteristics, manufacturing, inspection and test equipment calibration records, dimensions, marking, preservation, packaging, packing and operations, special and/or functional testing, necessary to assure compliance, and this information will be included in the FATR. The Contractor is responsible for providing the facilities and test equipment, and any additional administrative or technical support personnel necessary to assist in conducting the first article inspection and test. The product of this process is the FATR, which must include all of the information required by the First Article Test clause or Statement of Work contained in your contract. The importance of the test report cannot be underestimated, as it is the deliverable which forms the basis of acceptability of the First Article Test, notwithstanding the observations of Government Representatives while on-site. Do not hesitate to contact the AMRDEC Quality Assurance Specialist (QAS) through the contracting officer if you have questions regarding the content of the First Article Test Report.

Deficiencies. Deficiencies are defined as any nonconformance or departure from any contractually required material, physical, operational, functional or special characteristics, processes and/or procedures. The Contractor may, at its option, elect to submit a nonconforming first article as long as the appropriate waivers and/or deviations accompany the test report. There are risks associated with this decision, however, as the Government may either require a contract price adjustment, or elect to disapprove the FATR and require another first article test inspection at no additional cost.

Review of draft FATR. To increase the likelihood of a successful first time submission enclosure 1 contains a list of common causes for FATR rejection. To further increase the likelihood of a successful first time submission, if the contractor has a draft copy of the FATR, the AMRDEC QAS will perform a courtesy examination of it to see if anything that might cause a rejection can be identified and corrected prior to formal submission. This is not a guarantee that the FATR will be accepted, it is only a civility offered to increase that likelihood. Please note that if the procured item is highly complex, such as a transmission, and performing a courtesy review would significantly extend the length of the FAI trip that time may not permit a complete courtesy review.

Final Critique and Reporting. Upon completion of the first article inspection and test, a final meeting is held. Findings and/or results of the first article inspection and test are discussed and differences resolved. As the FATR is a contractual line item with inspection at origin, the test report must be annotated by the DCMA QAR recommending approval/disapproval. The Contractor will finalize the report and forward it through the Administrative Contracting Officer (ACO) to the Procurement Contracting Officer (PCO) who will provide it to Quality Management Division. In the event any first article inspection and test item is disapproved,

another first article inspection and test will be scheduled. (If the cause for disapproval was based solely on a simple issue such as a missing document that may be able to be resolved without a formal second inspection and test.) Any second inspection and test should be scheduled after the Contractor has implemented the appropriate actions to determine the cause of the nonconformance, taken required corrective actions, and presentation of reworked or replacement first article test piece. With PCO concurrence it is appropriate to contact the FAI QAS to seek clarity on FAI findings, but all official FAI/FATR corrective actions must be routed through the DCMA QAR to the PCO. Purchase of materials, components and /or commencement of production for the balance of the contract quantity is at the sole risk of the Contractor unless the PCO grants specific written authorization (see paragraph (g) of FAR 52-209-3 Alt 2).

SUMMARY

There may be variations in the basic outline presented herein, based on the peculiarities of specific products. The FAIs are thorough assessments to determine the ability of the Contractor to produce conforming material on a repetitive basis and to ascertain the adequacy of product specifications and operational requirements. When fully supported by Contractor and Government personnel, FAIs play an extremely important role in the overall objective to provide quality materiel to the Warfighter. Your cooperation and assistance in obtaining this objective is solicited and appreciated.

COMMON CAUSES OF FATR REJECTIONS

No indication of DCMA participation in first article test testing process (e.g. missing DD Form 1222 and/or DD Form 250)

No evidence that contractor has a Quality program as required by contract. (FAR 52.246-11)

Missing material/process certifications

Lack of traceability for material process/certifications showing requirements are flowed to subcontractors.

Missing special process certifications for personnel (e.g. NDT, Welding)

Wrong material used

A 100% dimensional characteristics check taken during first article test in-process and final inspections not recorded on appropriate documentation

No Part Marking or Bar Code Label Verification

Packaging/packing requirements not verified

Functional test not performed/documented as required

Lack of required calibration data

No certification the first article and production quantity will be produced in the same facility

Non-conforming characteristics not approved under waiver/deviation process

Lack of traceability for material process/certifications showing requirements are flowed to subcontractors.

NOTE: While no longer a cause for rejection, contractors can review MIL-HDBK-831 or AS9102 for examples of test report formats.